

Clifford Algebra Calibrations

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April 10, 2026

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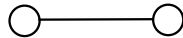
Simplex Geometry

$\emptyset$

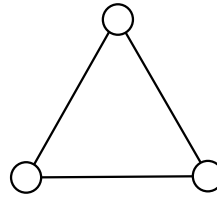
Null-simplex



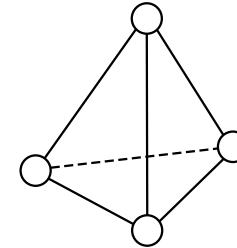
0-simplex



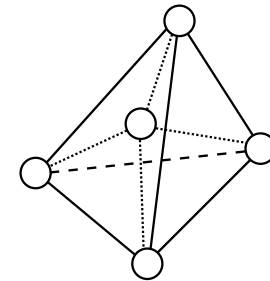
1-simplex



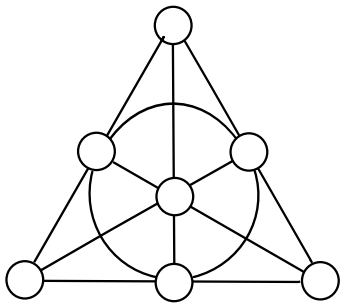
2-simplex



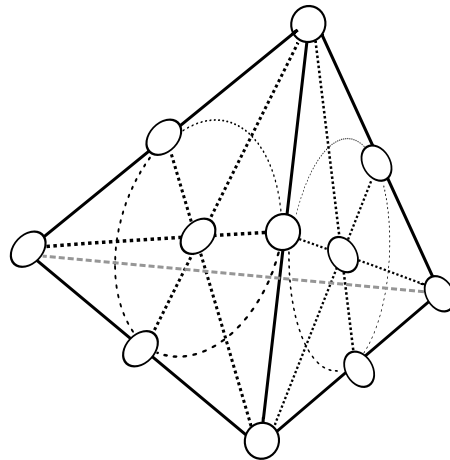
3-simplex



4-simplex



6-simplex



14-simplex

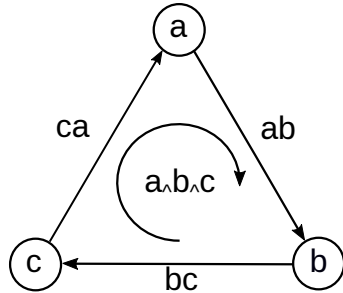
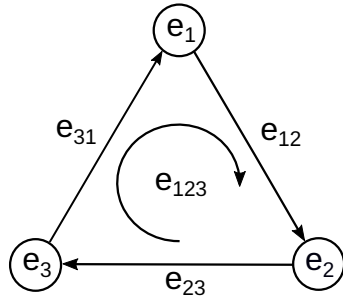
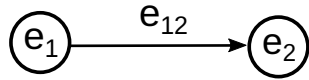
Fano plane is a projection of the 6-simplex with 21 edges and 35 faces displayed.

Fano volume is a projection of 15-D to 3-D with $15 \times 7 \times 3/3$ edges.

Pascal's Triangle

N	V	E	F	T	...			
1								
1	1							
1	2	1						
1	3	3	1					
1	4	6	4	1				
1	5	10	10	5	1			
1	6	15	20	15	6	1		
1	7	21	35	35	21	7	1	
			...					
1	15	105	455	1365	303	5005	6435	...
			...					...

Clifford Algebra



Rotation Algebra – 0° and 90° Basis

$$e_1 e_{12} = e_2, \quad e_2 e_{21} = e_1, \quad e_i e_j = \delta_{ij} + e_{ij}, \quad e_{ij} = e_i \wedge e_j$$

Full Rotation Group

$$ab = a \cdot b + a \wedge b = |a||b|(\cos \theta + \frac{a \wedge b}{|a||b_\perp|} \sin \theta)$$

$$a \times b = e_{321} a \wedge b \quad e_{321}^2 = -1$$

Left hand cross product given by $e_{123} = -e_{321}$

$$\begin{aligned} a_1 a_2 a_3 \dots a_n &= \sum_{i=0}^{\lfloor \frac{n}{2} \rfloor} \sum_{\mu \in \mathcal{C}_{2i}^n} (-1)^k \text{pf}(a_{\mu_1} \cdot a_{\mu_2}, \dots, a_{\mu_{2i-1}} \cdot a_{\mu_{2i}}) \\ &\quad a_{\mu_{2i+1}} \wedge \dots \wedge a_{\mu_n} \\ &= (a_1 a_2)(a_3 a_4) \dots \end{aligned}$$

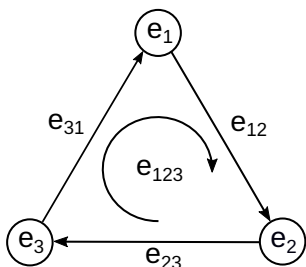
where $\mathcal{C}_{2i}^n$ is combinations $\binom{n}{2i}$ over the a_μ with parity k^1 .

Spin and Pin groups

Pfaffian of A , $\text{pf}(A)$, has the same properties as simplices. For $a_i^2 = 1$ get $\text{Spin}(n)$, n even, else odd $\text{Pin}(n) = \text{Spin}(n-1) + \text{a reflection}$, and these have $\leq \binom{n}{2}$ generalised Euler angle rotations. $\text{Pin}(n)$ has commuting pseudoscalar: $e_i e_{12\dots n} = e_{12\dots n} e_i$, $1 \leq i \leq n$, n odd.

¹G. P. Wilmot, The Structure of Clifford Algebra, *Journal of Mathematical Physics*, **28**, 1988

Sharp Algebras

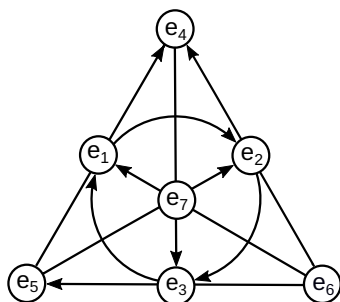


$$\text{Spin}^\sharp(3) = \{1, e_{12}, e_{13}, e_{23}\} \quad \text{Dual: } x^* = -e_{123}x$$

$$\text{Pin}^\sharp(3) = \{e_{123}, e_1, e_2, e_3\} + \text{Spin}^\sharp(3) = \text{Cl}(3) \cong \mathbb{C} \times \mathbb{H}$$

Projective geometry: vertices/edges shared twice

$$\Rightarrow e_i e_j = e_k^* \Rightarrow e_i^* e_j^* = e_k^*, \epsilon_{ijk} = 1 \Rightarrow \text{Spin}^\sharp(3) \text{ closed}$$

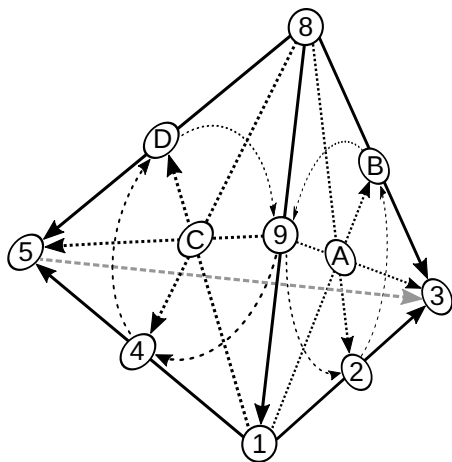


$$\text{Spin}^\sharp(7) = \{1, e_{1247}, e_{1256}, e_{1346}, e_{1357}, e_{2345}, e_{2367}, e_{4567}\}$$

$$\text{Pin}^\sharp(7) = -e_{1234567} \times \text{Spin}^\sharp(7) + \text{Spin}^\sharp(7)$$

$$\text{Dual: } x^* = -e_{1234567}x, \quad e_{1234567}^2 = -1$$

Projective geometry eg: $\Rightarrow e_{246}e_{356} \Rightarrow e_{167}^* \Rightarrow \text{Spin}^\sharp(7) \text{ closed}$



$$\text{Spin}^\sharp(15) = \{1, e_{1234567}^*, \Phi_{O_i}^*, \Phi_{P_{4i}}^*\}, \quad \text{for } 1 \leq i \leq 7$$

$$\text{Pin}^\sharp(15) = \{-1^*, e_{1234567}, \Phi_{O_i}, \Phi_{P_{4i}}\} + \text{Spin}^\sharp(15)$$

$$\Phi_O = \sum_{i=1}^7 \Phi_{O_i}, \quad \Phi_{P_4} = \sum_{i=1}^7 \Phi_{P_{4i}}$$

$$\text{Dual: } x^* = -e_{123456789ABCDEF}x, \quad e_{123456789ABCDEF}^2 = -1$$

Projective geometry eg: $\Rightarrow e_{1234567}e_{12389AB} \Rightarrow e_{123CDEF}^* \Rightarrow \text{Spin}^\sharp(15) \text{ closed, ad infinitum: } \text{Spin}^\sharp(31), \text{Spin}^\sharp(63), \dots$

Calibration $\text{Cl}(3)$

Definition A calibration is a p -form, ϕ , on a non-empty, closed, oriented, compact manifold, M , of $\mathbb{R}^n$ where

- (a) $d\phi = 0$ on M , and
- (b) $\phi_x(\xi) \leq 1$ for $x \in M$, $\phi_x \in \Lambda^p(\mathbb{R}^n)$ and $\xi = e_{\mu_1\mu_2\ldots\mu_p}$, $\mu_i \in \mathbb{N}_1^n$ and with same contact set then $\phi(\xi) = 1$.

This distinguishes a class of manifolds where $\phi(M) = 1$ is volume minimising

E.g. $\phi = e_{123}$, $\theta = \frac{1}{3}(x_1e_{23} + x_2e_{13} + x_3e_{12})$, $d\theta = \nabla\theta = \phi$, $d\phi = 0$, by Stoke's theroem

$$\int_{\partial M} \theta = \int_M d\theta = \int_M \phi = \int \phi(M)e_{123} = e_{123}$$

Fixed point $\text{Spin}^\sharp(3)$

$$\phi_3 = e_1 + e_2 + e_3, \quad \phi_3^* = e_{321}\phi_3 = -e_{12} + e_{13} - e_{23}$$

Invertible: $\phi_3^2 = 3$, $\phi_3^{*2} = -3$

Automorphisms: $\text{Spin}^\sharp(3)$ swaps pairs of quaternions so ϕ_3 and ϕ_3^* are invariant

Cl(3), Cl(0,3), Split Octonions, Octonions

$\text{Cl}(3)^2$	e_1	e_2	e_3	e_{23}	e_{13}	e_{12}	e_{123}
e_1	1	e_{12}	e_{13}	e_{123}	e_3	e_2	e_{23}
e_2	$-e_{12}$	1	e_{23}	e_3	$-e_{123}$	$-e_1$	$-e_{13}$
e_3	$-e_{13}$	$-e_{23}$	1	$-e_2$	$-e_1$	e_{123}	e_{12}
e_{23}	e_{123}	$-e_3$	e_2	-1	e_{12}	$-e_{13}$	$-e_1$
e_{13}	$-e_3$	$-e_{123}$	e_1	$-e_{12}$	-1	e_{23}	e_2
e_{12}	$-e_2$	e_1	e_{123}	e_{13}	$-e_{23}$	-1	$-e_3$
e_{123}	e_{23}	$-e_{13}$	e_{12}	$-e_1$	e_2	$-e_3$	-1

$\tilde{\mathcal{O}}^2$	u_1	u_2	u_3	u_{23}	u_{31}	u_{12}	u_{123}
u_1	1	u_{12}	$-u_{31}$	$-u_{123}$	$-u_3$	u_2	$-u_{23}$
u_2	$-u_{12}$	1	u_{23}	u_3	$-u_{123}$	$-u_1$	$-u_{31}$
u_3	u_{31}	$-u_{23}$	1	$-u_2$	u_1	$-u_{123}$	$-u_{12}$
u_{23}	u_{123}	$-u_3$	u_2	-1	u_{12}	$-u_{31}$	$-u_1$
u_{31}	u_3	u_{123}	$-u_1$	$-u_{12}$	-1	u_{23}	$-u_2$
u_{12}	$-u_2$	u_1	u_{123}	u_{31}	$-u_{23}$	-1	$-u_3$
u_{123}	u_{23}	u_{31}	u_{12}	u_1	u_2	u_3	1

$\text{Cl}(0, 3)^2$	i_1	i_2	i_3	i_{23}	i_{31}	i_{12}	i_{123}
i_1	-1	i_{12}	$-i_{31}$	i_{123}	i_3	$-i_2$	$-i_{23}$
i_2	$-i_{12}$	-1	i_{23}	$-i_3$	i_{123}	i_1	$-i_{31}$
i_3	i_{31}	$-i_{23}$	-1	i_2	$-i_1$	i_{123}	$-i_{12}$
i_{23}	i_{123}	i_3	$-i_2$	-1	i_{12}	$-i_{31}$	$-i_1$
i_{31}	$-i_3$	i_{123}	i_1	$-i_{12}$	-1	i_{23}	$-i_2$
i_{12}	i_2	$-i_1$	i_{123}	i_{31}	$-i_{23}$	-1	$-i_3$
i_{123}	$-i_{23}$	$-i_{31}$	$-i_{12}$	$-i_1$	$-i_2$	$-i_3$	1

$\mathcal{O}^2$	o_1	o_2	o_3	o_{23}	o_{13}	o_{12}	o_{123}
o_1	-1	o_{12}	o_{13}	$-o_{123}$	$-o_3$	$-o_2$	o_{23}
o_2	$-o_{12}$	-1	o_{23}	$-o_3$	o_{123}	o_1	$-o_{13}$
o_3	$-o_{13}$	$-o_{23}$	-1	o_2	o_1	$-o_{123}$	o_{12}
o_{23}	o_{123}	o_3	$-o_2$	-1	o_{12}	$-o_{13}$	$-o_1$
o_{13}	o_3	$-o_{123}$	$-o_1$	$-o_{12}$	-1	o_{23}	o_2
o_{12}	o_2	$-o_1$	o_{123}	o_{13}	$-o_{23}$	-1	$-o_3$
o_{123}	$-o_{23}$	o_{13}	$-o_{12}$	o_1	$-o_2$	o_3	-1

Maps to Cayley-Dickson algebras

Quaternions

$$e_{123} : \{(e_1, e_2, e_3) \rightarrow (o_1, o_2, o_{12})\}, \quad o_1 o_2 o_{12} = -1$$

Anti-quaternions

$$e_{321} : \{(e_1, e_2, e_3) \rightarrow (o_{12}, o_2, o_1)\}, \quad o_{12} o_2 o_1 = 1$$

Octonions

$$e_{1234567} : \{(e_1, e_2, e_3, e_4, e_5, e_6, e_7) \longrightarrow (o_1, o_2, o_{12}, o_3, o_{13}, o_{23}, -o_{123})\}$$

Sedenions

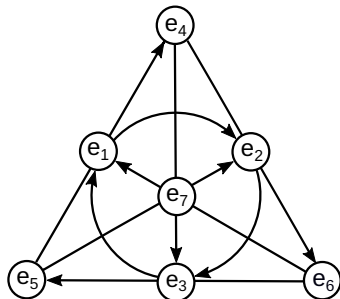
$$e_{123456789ABCDEF} : (e_1, e_2, \dots, e_F) \longrightarrow (o_1, o_2, o_{12}, o_3, o_{13}, o_{23}, o_{123}, o_4, \\ o_{14}, o_{24}, o_{124}, o_{34}, o_{134}, o_{234}, o_{1234})$$

Geometry $\rightarrow$ Graded notation for $Cl(n)$ – Grassmannian ordering (see Pascal's Triangle)

Doubling $\rightarrow$ Graded notation for Cayley-Dickson algebras
— ordering is derived from the binomial expansion

Quaternions in Clifford Algebra, $\{e_{ij}, e_{jk}, e_{ik}\}$, i, j, k unique, map to quaternions
or anti-quaternions in Cayley-Dickson elements, $o_\alpha o_\beta = \pm o_\alpha \text{ XOR } \beta$

Calibration Cl(7)



Associative calibration

$$\phi_7 = e_{123} + e_{145} + e_{167} - e_{246} + e_{257} + e_{347} + e_{356}$$

$$\phi_7^* = e_{1247} + e_{1256} + e_{1346} - e_{1357} + e_{2345} + e_{2367} + e_{4567}$$

Automorphisms: $\text{Spin}^\sharp(7)$ swaps three pairs of $\mathbb{H}$ so ϕ_7 and Φ_7^* are invariant, which provides a direct derivation of exceptional Lie algebra G_2 as double 90° rotations:

For e_{356}^* : $\frac{1}{2}(1 + e_{12})(1 \pm e_{47}) = \alpha + \beta$ so $(\alpha + \beta)\phi_7(\alpha - \beta) = \phi_7'$

where $\alpha = \frac{1}{2}(1 + e_{1247}), \beta = \frac{1}{2}(e_{12} + e_{47}), \alpha^2 = \alpha, \beta^2 = \alpha + 1, \alpha\beta = \beta\alpha = 0$

and importantly: $\beta\phi_7 = \phi_7'\beta$ and $\alpha\phi_7 = \alpha\phi_7'$ and $\phi_7' = \phi_7$ if $\beta \in G_2$.

Three rotations for each ϕ_7^* term: $e_{1247} \rightarrow \langle \beta = \left\{ \frac{1}{2}(e_{12} + e_{47}), \frac{1}{2}(e_{14} - e_{27}), \frac{1}{2}(e_{17} + e_{24}) \right\} \rangle$.

Permutation of the Fano plane: $7! = 30 \times 7 \times 3 \times 8$

i.e. 30 representations of seven 3-forms with 8 variations of signs or arrows.

Sign variations are parity of 3-forms and all $2^7 = 128$ sign variations of ϕ_7 gives $16 \mathbb{O}$.

Derivation of six power associative algebras: $\mathbb{P}_k, k \in \{4, 8, 10, 12, 14, 16\}$.

New Non-Associative Types

Associator typically assumes an “ordered” associator for $a < b < c$,

$$[a, b, c] = (ab)c - a(bc)$$

Unordered Associativity			Non-Associativity
$[b, a, c]$	$[a, b, c]$	$[a, c, b]$	
$\neq 0$	0	0	Type A
0	$\neq 0$	0	Type B
0	0	$\neq 0$	Type C
$\neq 0$	$\neq 0$	$\neq 0$	Type X

Simplifies Moufang equations and derives the structure of sedenions, trigintaduonions, etc.

$$\begin{array}{ccccccc}
 & & & \mathbb{P}_4 & \mathbb{P}_{12} & \mathbb{P}_{14} & \\
 & & & \downarrow & \downarrow & \downarrow & \\
 \mathbb{C} & \hookrightarrow & \mathbb{H} & \hookrightarrow & \mathbb{O} & \hookrightarrow & \mathbb{U}_1 \hookrightarrow \mathbb{U}_2 \hookrightarrow \mathbb{U}_3 \hookrightarrow \mathbb{U}_4 \hookrightarrow \dots
 \end{array}$$

Sedenions have 8 octonions and 7 $\mathbb{P}_4$ subalgebras [Cawagas 2004], hence 15 Fano planes.

Calibrations in $Cl(15)$

15 dimensional so e_F and $e_{123456789ABCDEF}$ are in $Cl(15)$ with cross product operator

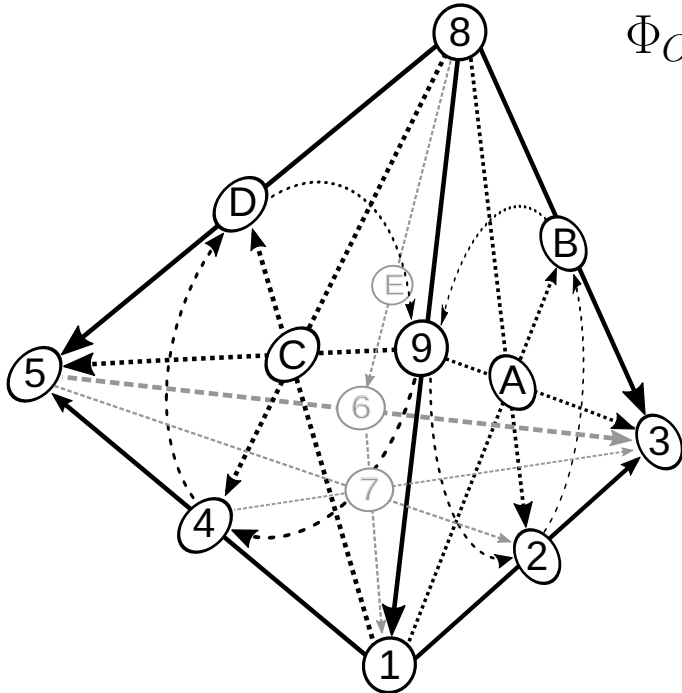
$$\begin{aligned}\phi_{15} = & e_{123} + e_{145} + e_{167} + e_{189} + e_{1AB} + e_{1CD} + e_{1EF} + e_{246} + e_{257} + e_{28A} + e_{29B} + e_{2CE} \\ & + e_{2DF} + e_{347} + e_{356} + e_{38B} + e_{39A} + e_{3CF} + e_{3DE} - e_{48C} + e_{49D} + e_{4AE} + e_{4BF} + e_{58D} \\ & + e_{59C} + e_{5AF} + e_{5BE} + e_{68E} + e_{69F} + e_{6AC} + e_{6BD} + e_{78F} + e_{79E} + e_{7AD} + e_{7BC}\end{aligned}$$

This associative calibration maps to a sedenion representation but is not invertible

The 8 $\mathbb{O}$ and 7 $\mathbb{P}_4$ form a new calibration:

$$\Phi = \Phi_A + \Phi_O + \Phi_P, \quad \Phi_A = e_{1234567}, \quad \Phi_P = e_{89ABCDEF}\Phi_O,$$

$$\begin{aligned}\Phi_O = & e_{12389AB} + e_{14589CD} + e_{16789EF} + e_{2468ACE} \\ & + e_{2468ACE} + e_{2578ADF} + e_{3478BCF} + e_{3568BDE}.\end{aligned}$$



Dual Φ^* terms are in $Spin^\sharp(15)$

Each swaps 7 octonion-like pairs in the calibration

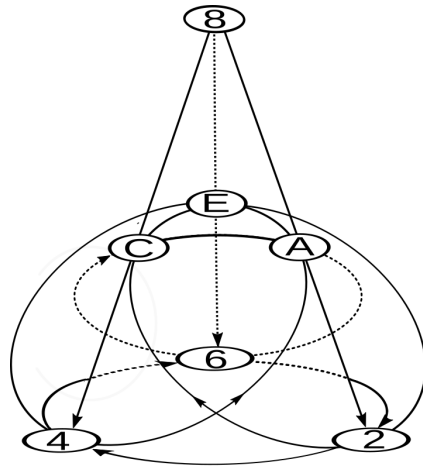
This is the Fano volume

Sharp(3) is a 1-D projection (o-o-o)

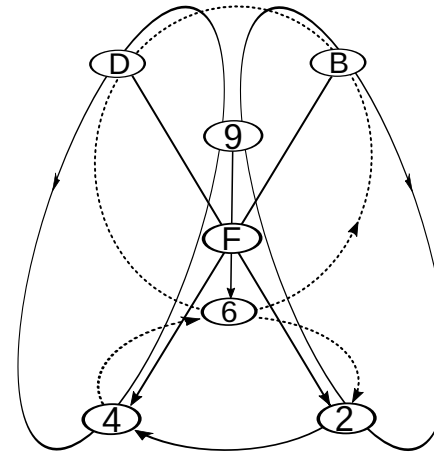
Sharp(31) projects to a 4-D Fano-hyper-volume

Sharp(63) projects to a 5-D Fano-hyper-volume

Fano Cone and Folly



(a) $\mathbb{O}$ Fano Cone



(b) $\mathbb{P}_4$ Fano Folly

Physics

Furey has some success applying $\mathbb{O}$ to particle physics ($SU(3) \subset G_2$)

Gresnight suggests 3-colour fermions as 3 $\mathbb{O}$ in $\mathbb{S}$ with shared $\mathbb{H}$

To avoid $\mathbb{P}_4$ need edges that include e_8 ($e_{15} + e_{25} + e_{35}$ or $e_{123} + e_1 \dots$ or $e_{123} + e_3 \dots$)

Instead of octonions, U_3 has $\mathbb{P}_4, \mathbb{P}_{12}$ and $\mathbb{P}_{14}$, which could have a family of G_2 's

Then look for 3 generations of lepton and 3-colour fermion particle families

Automorphisms in $\text{Cl}(15)$

Each dual term derives 28 rotations, e.g. $e_{3568\text{BDE}}^* = e_{12479\text{ACF}} \rightarrow \frac{1}{4}(e_{12} \pm e_{47} + e_{9\text{A}} \pm e_{\text{CF}})$

The fifteen 8-form generate 3 Cyclic and 4 Mixed rotations with 4 sign variations each

$$\Phi^{C(1+2+3+4)} \text{ and } \Phi^{M(1+2+3+4)}$$

But only the $\Phi_O^{C(1+2+3+4)}$ and $\Phi_A^{C\&M(1+2+3+4)}$ parts form closed algebras below

$\Phi_O^{C(1)}$ is a 15-D representation of octonions and $\Phi_O^{C(1+2)} \cong G_2 \otimes \mathbb{C}$

$\Phi_O^{C(1+2+3+4)}$ can't be either direct product $G_2 \otimes D_4$ or $G_2 \otimes K_4$

$\Phi_O^{C(\alpha)} \wedge \Phi_O^{C(\beta)}$				
$\alpha \backslash \beta$	1	2	$3 \rightarrow (1+3)$	$4 \rightarrow (2+4)$
1	$\Phi_O^{C(1)}$	$\Phi_O^{C(2)}$	$\Phi_O^{C(1+3)}$	$\Phi_O^{C(2+4)}$
2	$\Phi_O^{C(2)}$	$\Phi_O^{C(1)}$	$\Phi_O^{C(2+4)}$	$\Phi_O^{C(1+3)}$
$3 \rightarrow (1+3)$	$\Phi_O^{C(1+3)}$	$\Phi_O^{C(2+4)}$	$\Phi_O^{C(1+3)}$	$\Phi_O^{C(2+4)}$
$4 \rightarrow (2+4)$	$\Phi_O^{C(2+4)}$	$\Phi_O^{C(1+3)}$	$\Phi_O^{C(2+4)}$	$\Phi_O^{C(1+3)}$

Automorphism Lie Products (continued)

$\Phi_A^{C\&M(\alpha)} \wedge \Phi_A^{C\&M(\beta)}$				
$\alpha \backslash \beta$	1	2	3	4
1	$\Phi_A^{C\&M(1+2)}$	$\Phi_A^{C\&M(1+2+3)}$	$\Phi_A^{C\&M(2+3)}$	$\Phi_A^{C\&M(4)}$
2	$\Phi_A^{C\&M(1+2+3)}$	$\Phi_A^{C\&M(1+2)}$	$\Phi_A^{C\&M(2+3)}$	$\Phi_A^{C\&M(4)}$
3	$\Phi_A^{C\&M(2+3)}$	$\Phi_A^{C\&M(2+3)}$	$\Phi_A^{C\&M(1+2+3)}$	$\Phi_A^{C\&M(4)}$
4	$\Phi_A^{C\&M(4)}$	$\Phi_A^{C\&M(4)}$	$\Phi_A^{C\&M(4)}$	$\Phi_A^{C\&M(1+2+3)}$

The change of basis to $\Phi_O^{C(1+3)}$ and $\Phi_O^{C(2+4)}$ changes Φ_O^C to a semi-direct product

A change of basis of Φ_A^A simplifies it to $\Phi_A^{A(1+2+3)} \oplus \Phi_A^{A(4)}$

But which form the automorphisms of Sharp(15)?

Analogous to Cl(7), the full rotation is $x' = (\alpha + \beta)x(\alpha - \beta)$, where $x \in \text{Sharp}(15)$'s basis, α is 0, 4 and 8-forms that all commute with x and β is the 2 and 6-forms

Only $\Phi_O^{C(1)}$ keeps x invariant (like 7-D G_2) and all other invariants permute the x 's

Furthermore, α is a stabiliser, $\alpha x = \alpha x'$, only for $\Phi_O^{C(1)} \implies \Phi_O^{C(1)} = \text{Aut}(\text{Sharp}(15))$

Automorphisms in $\mathbb{S}$

All 420 invariants of Φ are not closed under multiplication

To map to $\mathbb{S}$ the associative calibration, ϕ_{15} , must be invariant

For $\mathbb{O}$, G_2 are invariants and for $\mathbb{S}$, $\Phi_O^{C(1)}$ are invariants (i.e. G_2)

All other $\Phi_O^{C(2+3+4)}$, Φ_P^C , $\Phi_A^{C\&M}$ rotate Φ_{15} to other $\mathbb{S}$ representations

While Φ_O^M and Φ_P^M rotate to quasi-sedenions

Separate domains again and $\Phi_P^{C(3+4)}$ has the same issue as $\text{Spin}^\sharp(15)$.

The full rotation argument of Sharp(15) applies to Φ and the 3-form calibration

$$(\alpha + \beta)\phi_{15}(\alpha - \beta) = \phi'_{15}$$

with $\beta\phi_{15} = \phi'_{15}\beta$ and $\alpha\phi_{15} = \alpha\phi'_{15}$ for $\Phi_O^{C(1)}$ only, so $\alpha\phi_{15}$ is a stabiliser for $\Phi_O^{C(1)} \cong G_2$

Schafer proves $\text{Aut}(\mathbb{S}) = G_2$ and Brown and Eakin and Satheye prove $\text{Aut}(\mathbb{S}) = G_2 \times S_3$

Hence I support Schafer's result.

Results

Clifford algebra has calibrations for many Lie algebras plus the Cayley-Dickson algebras which provides tools for analysis and a standard mathematical framework.

Invertible Clifford group, Cl^* , is more than just the Spin and Pin groups but also calibrations.

This work was verified with Python calculators at <https://github.com/GPWilmot/geoalg>.

Open Questions

My conjecture is that the Invertible Clifford group, Cl^* , equals calibrations.

Clifford algebra should have calibrations for the Symmetric group (including the Monster²).

Are there calibrations for algebras not currently known, such as Φ_O^C ?

Acknowledgements

I wish to acknowledge my supervisors for guidance and useful discussions.

²The Monster is the largest group (196,883 dimensions) and is in the sporadic groups.